



Research Proposal Approval Format

As per the guideline of Nepal Health Research Council (NHRC)

Research Title:

TO STUDY ABOUT CORTICOSTEROIDS RELATED PRESCRIPTION IN MMTH

NEHCO-IRC

Manmohan Memorial Institute of Health Science (MMIHS)

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For Official Use Only

(Please see the check list before Registration of the application form)

Registration No.:

Registration Date:

Approved Date:

Name of PI:

Total Budget of the Project:

NHRC Processing Fee:

Research site:

Tentative date of Initiating the Project:

Duration of the Research Project:

Name of Internal Reviewer:

Name of External Reviewer:

Signature & Seal of NHRC:

Part – I

Passport size
photograph

Administrative Information

1. Research Title:

STUDY OF DISSOLUTION ENHANCEMENT OF ACECLOFENAC TABLETS BY
SOLID DISPERSION TECHNIQUE

2. Name and Title of Principal Investigator responsible for the proposed research:

Last (Surname)

Middle (if any)

First name

Nationality:

Citizenship Number with district name from where it was obtained (only for Nepali):

Passport Number (only for non-Nepali citizen):

Signature:

Date:

Postal Address:

Telephone no.:

Mobile No.:

Fax No.:

e-mail:

Alternate e-mail:

3. Full name of the Institution associated with the Principal Investigator

(if applicable):

Designation:

Postal Address (if different from the address given above):

Telephone No.:

Fax No.:

e-mail:

Website:

4. Declaration of the head of the Institution (if applicable)

If the proposed research is approved, we will allow him/her to conduct the research in this institution.

Signature: Date:

Last (Surname) Middle (if any) First name

Designation:

Name of the Institution

Contact/Postal Address:

Telephone No.:

Fax No.:

Institutional e-mail:

Website:

5. Name and Title of Co-investigators responsible for the proposed research

(Use the similar format if more than one):

-
- Last (Surname) Middle (if any) First name

Nationality:

Citizenship Number with district name from where it was obtained (only for Nepali)

Passport Number (only for non-Nepali citizen):

Affiliated Institution (if applicable):

Manmohan Memorial Institute of Health Sciences, Kathmandu

Designation: Student, B.Pharmacy 4th year

Signature:

Date:

Postal Address (if different from the address given above):

N/A

Telephone No.: 9865209008

Fax No.: N/A

e-mail: gtmroshan4u@gmail.com

(Use additional sheet if necessary)

6. List the name(s) and institutional affiliation to the researcher(s) (other than co-investigator) to assist your project in Nepal and abroad (if any)

Name

Institution and Address

(a) MMIHS, Kathmandu

(b) MMIHS, Kathmandu

(Use additional sheet if necessary)

7. List the name(s) of Nepali researcher(s) (other than co-investigator) or Nepalese Institution/hospital/NGO(s) etc. from whom you may seek co- operation (if any)

(a) Manmohan Institute of Health Sciences

(Use additional sheet if necessary)

8. List major equipment(s) in relation to your research project you plan to bring/import to Nepal (If applicable)

(a)

(b)

(Use additional sheet if necessary)

8.1 List details of all specimen(s) (if any) that you may transport from Nepal in relation to your research.

(a)

(b)

(c)

(d)

8.2 Country of Destination:

Name of Institution:

8.3 Mode of Transportation of Specimen

8.4 How will you ensure duplicate specimens remain in the country?

(If necessary, use additional sheet)

9. Is this research part of your thesis?

Yes ☒

No ☐

If yes,

For what degree and in which subject?

From which university?

From which country?

Part – II**Financial Information**

9. Research Title:

STUDY OF DISSOLUTION ENHANCEMENT OF ACECLOFENAC TABLETS BY
SOLID DISPERSION TECHNIQUE

10. Name of the funding organization:

Self-funded

Contact information of funding organization or agency:

Postal Address:

N/A

Telephone No.:

N/A

Fax No.:

N/A

e-mail:

N/A

Contact person at the funding organization or agency:

Last (Surname)

Middle (if any)

First name

Designation:

Total amount of funds (in NRs / US \$) allocated for the proposed research
project:

Itemized budget (in detail) and justify the resources required for the proposed
research work (*use additional sheet*)

Part – III

Research Proposal Description

11. Research Title:

STUDY OF DISSOLUTION ENHANCEMENT OF ACECLOFENAC TABLETS BY SOLID DISPERSION TECHNIQUE

12. Proposal Summary (maximum 500 words):

Aceclofenac is a widely used non-steroidal anti-inflammatory drug (NSAID) known for its effective analgesic, anti-inflammatory, and antipyretic properties. It is commonly prescribed for the treatment of osteoarthritis, rheumatoid arthritis, and ankylosing spondylitis due to its ability to provide prolonged pain relief with fewer gastrointestinal side effects compared to other NSAIDs. Despite its therapeutic advantages, aceclofenac belongs to the Biopharmaceutics Classification System (BCS) Class II, characterized by high permeability but low aqueous solubility. This poor solubility leads to limited dissolution in the gastrointestinal tract, which in turn results in reduced oral bioavailability and delayed onset of action.
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To overcome this limitation, this research focuses on enhancing the dissolution rate of aceclofenac using the solid dispersion technique. Solid dispersion is a proven strategy for improving the solubility of poorly water-soluble drugs by dispersing them in a hydrophilic polymer matrix. By increasing the drug's surface area and reducing crystallinity, this method promotes faster and more complete dissolution. The primary aim of this study is to improve aceclofenac's solubility and bioavailability, thereby enhancing its therapeutic effectiveness. The successful formulation of a solid dispersion system could potentially lead to more efficient drug delivery and better patient outcomes.

13. Introduction:

13.1 Background of Study (maximum 500 words):

Acetoclofenac is an oral non-steroidal anti-inflammatory drug (NSAID) with antiinflammatory and analgesic properties. Acetoclofenac is a phenylacetic acid derivative nonsteroidal anti-inflammatory drug (NSAID) with marked anti-inflammatory and analgesic properties.^{1,2} It is a potent inhibitor of cyclooxygenase (COX),^{1,3,4} a key enzyme in the synthesis of prostaglandins and thromboxanes with selectivity for the COX-2 over COX-1 isoform.¹ Acetoclofenac is mainly recommended for the treatment of inflammatory and painful processes, such as low back pain, scapulohumeral periarthritis, extraarticular rheumatism, odontalgia, and osteoarthritis, rheumatoid arthritis (RA), and ankylosing spondylitis. Acetoclofenac is at least as effective as other NSAIDs in reducing pain and/or improving functional capacity in chronic pain conditions (OA, AS, RA, and LBP). It is generally well tolerated and appears to have a more favorable GI profile than other NSAIDs. Thus, current evidence indicates that acetoclofenac is a useful option for the management of pain and inflammation across a wide range of painful conditions.

13.2 Statement of the Problem and Rationale / Justification (maximum 500 words)

Statement of the Problem:

Solubility and Lipophilicity are two of the most important Rate limiting factor of pharmacokinetics . According to these two factors, A classification was introduced which is called as Biopharmaceutics classification System. The Biopharmaceutics Classification System (BCS) categorizes drug molecules into four groups based on their solubility and Lipophilicity (permeability) profiles:

- Class I: high permeability, high solubility compounds
- Class II: high permeability, low solubility compounds
- Class III: low permeability, high solubility compounds
- Class IV: low permeability, low solubility compounds

Up to 50% of all the authorized drug are categorized in classes II and IV . For molecules belonging to classes II and IV, the main goal in formulation development is increasing

their solubility. Aceclofenac is also BCS class II drug as it has high permeability but has low solubility. If a drug has low solubility, it has low dissolution rate and as consequence, the bioavailability will also be less. It results in low therapeutic value of the drug. The drug aceclofenac is a newly researched and developed compound. So, if It gets upgraded, it will be more competent in mitigating the diseases.

Rationale

Enhancing the dissolution profile of BCS class II drugs like Aceclofenac is crucial to ensure consistent therapeutic efficacy and predictable pharmacokinetics. One of the most promising approaches to overcome this limitation is the **solid dispersion technique**, which involves dispersing the drug in an inert carrier or matrix in the solid state. This method can improve the wettability, reduce particle size, and convert the drug into an amorphous form- factors that collectively enhance dissolution.

This study aims to formulate and evaluate aceclofenac solid dispersions using various hydrophilic carriers to improve its dissolution profile. By employing the solid dispersion technique, we can address the solubility-related challenges of aceclofenac, thereby potentially enhancing its therapeutic efficacy. This study also aims to improve patient compliance by offering faster and more effective pain relief.

13.3 Research Objectives / purpose / aim of the study:

General

Study of Dissolution enhancement of Aceclofenac Tablets by Solid Dispersion Technique.
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Specific

- | |
|---|
| <ul style="list-style-type: none">• To prepare solid dispersions of aceclofenac tablets using hydrophilic carriers• To formulate aceclofenac tablets using the optimized solid dispersion• To evaluate and compare the dissolution profile of the formulated tablets with conventional or Marketed Aceclofenac tablets. |
|---|

14. Research Design and

Methodology Research Method

Qualitative ☐ Quantitative ☐ Combined ☒

Study Variables:

Dependent Variables

- Type of hydrophilic carrier (Means polymer used)
- Drug-to-polymer ratio
- Method of preparation (e.g., solvent evaporation, melting method)
- Processing parameters (temperature, mixing time etc also determines)

Independent Variables

- Dissolution rate of Aceclofenac
- Drug content and uniformity
- Tablet disintegration time
- In-vitro release profile
- Stability of the formulation

Type of Study (Specify):

Experimental and Observational.

Study Site and Its Justification:

Laboratory of Manmohan Memorial Institute of Health Sciences

Study Population (Specify):

Study Unit:

Aceclofenac

Sampling Methods / Techniques (Specify):

Sample size (with justification):

Criteria for Sample Selection:

Data Collection Technique / Methods (Specify):

Materials:

The research demands Acclofenac, HPMC, PVP-K30, Polysorbate 80, MCC, crospovidone, Talc, Magnesium stearate.

Data Collection Tools: (please attached in annex)

Pre-testing the Data Collection Tools (if applicable):

Validity and Reliability of the Study Tools:

All validated protocols

Potential Biases (if applicable):

N/A

Limitation of the Study:**15. Plan for Supervision and Monitoring:**

Regular contact with supervisors, seeking advice and informing about the progress will be done via email, calls or messages. The supervisor will be informed about the progress during the process of data collection that is from the start to the end of data collection.

Plan for Data Management and Analysis:

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16. Expected Outcome of the Research:

17. Plan for Dissemination of Research Results:

The final report will be submitted to Research Department, Manmohan Memorial Institute of Health Sciences. Results will be presented in conferences of the department and will be published in research report.

18. Plan for Utilization of the Research Findings (optional):

Research outcome will be publishing in research journal.
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How is the research project going to strengthen the research capability of the host institution: Nepali Researcher (if submitted from aboard):

N/A

19. Work Plan (*should include duration of study, tentative date of starting the project and work schedule / Gantt chart*):

S.N.	Task (2024/25)	May	Jun	Jul	Nov	Feb	Mar	Apr	May	June	july
1	Literature review										
2	Topic selection										
3	Topic presentation										
4	Proposal writing										
5	Proposal submission										
6	Proposal approved by IRC										
7	Data collection										
8	Data entry and analysis										
9	Report writing										
10	Report presentation and submission.										

Part – IV

Ethical Consideration

20. Regarding the human participants:

Are human participants required in this research? If yes, provide justification.

☐

Yes (*provide justification*)

No

☐

How many participants are required for the research? Explain.

What is the frequency of the participant's involvement in the research? Explain.

Clearly indicate the participant's responsibilities in the research. What is expected of the research participants during the research?

Are vulnerable members of the population required for this research? If yes, provide justification.

Are there any risks involved for the participants? If yes, identify clearly what are the expected risks for the human participants in the research and provide a justification for these risks.

Are there any benefits involved for the participants? If yes, identify clearly what are the expected benefits for the participants.

1. Informed Consent Form / Ethical Issues:

Statements required in the Informed Consent Form include:

A statement that the human participants can withdraw from the study at any time without giving reason and without fear. State clearly how the participants can opt out the study.

A statement guaranteeing the confidentiality of the research participants.

If required, a statement on any compensation that might be given to the research participant and or their community.

A statement indicating that the participants has understood all the information in the consent form and is willing to volunteer / participate in the research.

Signature space for the research participants, a witness, and the date.

(Informed Consent form should be submitted in English and in the language appropriate to the research participants)

Obtaining the Consent

How informed consent is obtained from the research participants?

☐

Verbal

☒

Written

Please indicate who is responsible for obtaining informed consent from the participants in this research study?

Researcher will be responsible for obtaining informed consent from the participants in this research study.

Is there anything being withheld from the research participants at the time the informed consent is being sought?

If yes, explain

...No.....

Is the research sensitive to the Nepali culture and the social values?

Yes ☐

No ☒

Explain.

.....

Is health insurance (*if applicable*) being made available to the research participants? If yes, please provide the necessary insurance data.

.....

(Include in consent form)

2. Regarding Clinical Trial:

In case of a clinical trial address the following:

The trial treatment

.....

A detailed explanation of the trial procedures including all invasive procedures.

.....

The potential or direct benefits (if any) for the research participants.

.....

Alternative procedure(s) or treatment(s) that may be available.

.....

The risks, discomforts, and inconveniences associated with the study

.....

Provisions for management of any adverse reactions

.....

The provisions of insurance coverage for any permanent disability or death caused directly by the investigational treatment or procedure.

.....

The provision of including the name and address, including telephone numbers of person to be contacted in case of adverse events or for any information related to the trial.

.....

Is there going to be a transfer of any biological materials from the country? Explain.

.....

Is there a Data Safety Monitoring Board?

If Yes, Mention

.....

Is this trial internationally registered?

.....

Part – V

ACCEPTANCE OF GENERAL CONDITIONS AND DECLARATION BY THE PRINCIPAL INVESTIGATOR

I hereby certify that the above-mentioned statements are true, I have read and understood the regulation of the Nepal Health Research Council (NHRC) on the approval of research proposal and will act in conformity with the said regulation in all respects.

If the research is terminated, for any reason, I will notify NHRC of this decision and provide the reasons for such actions. I will provide NHRC with a written notice upon the completion of the research as well as a final summary/full report of the research study. If I publish the results in a journal, I shall acknowledge the NHRC and shall provide the Council with three copies of any such articles.

.....

Signature of Applicant

Date:

INFORMED CONSENT:

- Describe the manner in which informed consent will be obtained.
- Indicate what kind of consent (e.g. parental, child, adult, etc) will be used.
- If the subjects are children/adolescents ages 7-18 years, an Assent Form must be included with the IRB application. The signed Assent Form along with the Parental/Guardian Consent Form must be retained on file for at least three years after completion of the research project.
- If prisoners / pregnant women, or fetuses are to be included in the research sample, it is likely that a full IRB review will be required and additional human subjects' protections will be expected.
- If the subjects do not read or comprehend English, you must provide a consent form in their language as well as in English for IRB review and approval.
- If you are requesting a waiver of written consent (i.e. a signature on an informed consent form) from the subjects, you **MUST** justify this request by providing an explanation of why obtaining written consent would add additional risk to the subjects and your alternative provisions for informing them about the study.
- If consent documents from another site will be used, you will have to indicate this and provide a copy of the authorized consent document and IRB approval with your application.
- You will have to provide any other relevant information if necessary. Please be aware that the PI is legally required to retain all signed Informed Consent forms for at least three years after the project terminates
- The Informed Consent form must be written at a level that the subjects will understand.
Please use simple language, and avoid clinical jargon.

- Attach a copy of the written informed consent form (assent or parental consent where applicable). Consent documents **MUST** be in format requested. See examples on line.
- If the study uses database or archival data the use of informed consent is not applicable.

CONFIDENTIALITY OF DATA: *Confidentiality of data MUST be address for all studies.*

- Indicate the extent to which confidentiality of records identifying subjects will be maintained.
- Describe the storage and disposal of information where applicable.

Check List

For all applicants

1. Covering letter addressed to the Member secretary indicating the submission of the approval of proposal.
2. Proposal will only be accepted if submitted in NHRC format.
3. Both printed and electronic version of the proposal should be submitted.
4. Curriculum Vitae of the Principal Investigator & Co-Principal Investigator of the study team should be submitted.
5. If the Principal Investigator is a non Nepali citizen, at least one Co-investigator should be a Nepali citizen.
6. Submission of the application processing fee to NHRC
(Below US \$ 10,000 is US \$100.00 and above US \$ 10,000 is 3% of the total budget).
7. Source of funding for the proposed project.
8. The proposal should have institutional ethical clearance from his/her own country if submitted from academic and related institution.
9. If the research study is to be conducted in any hospitals/organization or institution/community, a letter of approval from the related hospital/organization or institution/district authority should be provided.
10. Consent form should be in Nepali & local language (if necessary).
11. Data collection tools should be in Nepali & local language (if necessary) including interview guideline, observation checklist, questionnaires etc.
12. Style of referencing should be in Harvard style.
13. List of abbreviations / acronyms should be provided.

For students' applicants

1. Approval letter from concern Institute/University.
2. Recommendation letter from Academic Supervisor.

Circular for Researcher

NHRC has decided following fees for the research proposals that are submitted for approval from NHRC with effect from 2066/11/27 (11th March 2010).

S.N.	Budget in the research/thesis proposal	Minimum processing fees
1.	Nepali Student (Thesis Proposal) studying in Nepal	NRs. 1,000.00
2.	Nepali Student (Thesis Proposal) studying in abroad	US \$ 100.00
3.	Foreigner Student (Thesis Proposal)	US \$ 100.00
4.	Research Activities supported by NHRC Account	15% of the Total Budget
5.	Research Project (Total Budget up to US \$ 10,000.00)	US \$ 100.00
6.	Research Project (Total Budget above US \$ 10,000.00)	3% of the Total Budget

References

1. Iolascon, G., Giménez, S., & Mogyorósi, D. (n.d.). *A review of Aceclofenac: Analgesic and anti-inflammatory effects on musculoskeletal disorders*. University of Campania “Luigi Vanvitelli”; UGC Limonar; State Medical Center of Szentendre; Buda Health Center.
2. Huang, B. B., Liu, D. X., Liu, D. K., & Wu, G. (n.d.). *Application of solid dispersion technique to improve solubility and sustain release of emamectin benzoate*.
3. Chitosan and Enteric Polymer Based Once Daily Sustained Release Tablets of Aceclofenac: In Vitro and In Vivo Studies Srinivas Mutalik,^{1,3,4} Krishnan Manoj,¹ Meka Sreenivasa Reddy,¹ Pralhad Kushtagi,² Achutha Nayak Usha,¹ Parambil Anju,¹ Averineni Kumar Ranjith,¹ and Nayanabhirama Udupa¹
4. Kuppusamy, G., Singh, S. K., Dahiya, D. P., & [4th Author's Name]. (n.d.). *Dissolution enhancement of poorly soluble Aceclofenac by solid dispersion technique and its comparison with marketed formulations*. *International Journal of PharmTech Research*.
5. Chowdhury, R., Anjum, R., Siddiqui, S. A., Rahman, M. Z., Sharmin, T., & Rashid, M. M. O. (n.d.). *Formulation and evaluation of Aceclofenac solid dispersion for improving the solubility and dissolution rate*. Department of Pharmacy, Noakhali Science and Technology University.
6. Choi, J.-S., Lee, S.-E., Jang, W. S., Byeon, J. C., & Park, J.-S. (n.d.). *Solid dispersion of dutasteride using the solvent evaporation method: Approaches to improve dissolution rate and oral bioavailability in rats*. College of Pharmacy and Institute of Drug Research and Development, Chungnam National University.
7. Sarangi, M. K., & Singh, N. (n.d.). *A comparative study of solubility enhancement of Aceclofenac by solid dispersion technique using several polymers*. Sardar Bhagwan Singh PG Institute of Biomedical Sciences and Research.
8. Shaik, N. B., Toleti, M., Kanaru, D., & Kukati, L. (n.d.). *Solubility enhancement of Aceclofenac by solid dispersion*. G. Pulla Reddy College of Pharmacy, Mehdipatnam, Hyderabad, Telangana, India.
9. Neupane, S., & Thapa, C. (n.d.). *Formulation and enhancement of dissolution rate of poorly aqueous soluble drug Aceclofenac by solid dispersion method: In vitro study*. Department of Pharmacy, Universal College of Medical Sciences, Tribhuvan University, Siddhartha Nagar, Rupandehi, Nepal.

Annex 1: CURRICULUM VITAE

CURRICULUM VITAE

Roshan Gautam

Date of Birth: 25/10/2002

Nationality: Nepali

Gender: Male

Present Address: Satungal, Kathmandu, Nepal

Permanent Address: Bardaghat, Nawalparasi, Nepal

Email: gtmroshan4U@gmail.com

Contact number: 9865209008

ACADEMIC QUALIFICATION

Date	Academic Institute/University/country	Degrees
2024	Manmohan Memorial Institute of Health Sciences/TU/Nepal	Bachelor- 4 th year Running
2019	Aroma College	+2 Sciences
2017	Sun Rise English School	SEE

TRAININGS

Date	Name of Training	Organizers
2023	Academic Proposal Development Training	MMIHS, MPSA, SQC
2023	GMP Training Program	NPSS
2022	First Aid Training	MPSA
2022	Modern respiratory drug delivery system training	CIPLA, MPSA

REFERENCES:

- Prof. Dr. Dharma Prasad Khanal

Campus chief of Manmohan Memorial Institute of Health sciences, Solteemode, Kathmandu (Affiliated to Tribhuvan University) Email: drdharmakhanal@gmail.com

- Ass. Prof. Mijala Bajracharya

Head of Department of Pharmacy, Manmohan Memorial Institute of Health sciences,
Solteemode, Kathmandu Email: bmijala@yahoo.com